

Project Configuration Report

doan_final

Target Device:

Target Part: EFR32MG21A020F1024IM32

Boards:

- Wireless Starter Kit Mainboard (BRD4001A Rev A01)
- EFR32xG21 2.4 GHz 20 dBm Radio Board (BRD4180B Rev A03)

Target SDK

Simplicity SDK Suite v2024.6.2: Amazon 202012.00, Bluetooth 8.2.0, Bluetooth Mesh 7.0.2, EmberZNet 8.0.2.0, Flex 3.8.2.0, Micrium OS Kernel 5.16.00, OpenThread 2.5.2.0 (GitHub-1fceb225b), Platform 5.0.2.0, Silicon Labs Matter 2.3.2-1.3, USB 1.3.1.0, Wi-SUN 2.2.0.0, Z-Wave SDK 7.22.2.0
SDK Version: 2024.6.2

Extensions

- Silicon Labs Matter 2.3.2

Project Generators

- Simplicity IDE Project

Installed Software Components:

- Application

- Utility

- Assert**

- Configurable: true

- Configs: app_assert_config.h

- Log**

- Sources: app_log.c

- Configurable: true

- Configs: app_log_config.h

- Bluetooth

- Application

- Firmware Update

- In-Place OTA DFU**

- Sources: sl_bt_in_place_ota_dfu.c

- Configurable: true

- Configs: in_place_ota_dfu.xml, sl_bt_in_place_ota_dfu_config.h

- GATT Services

- Device Information GATT Service**

- Sources: sl_gatt_service_device_information.c

- Configurable: false

- Static GATT Database and Configuration**

- Configurable: true

- Configs: gatt_configuration.btconf

- Miscellaneous

- BLE Post Build**

- Configurable: false

- **Bluetooth Host (Stack)**

- Bluetooth Core**

- Sources: sl_bt_stack_init.c

- Configurable: true

- Configs: sl_bluetooth_config.h

- Features with Commands and Events

- GATT Client**

- Configurable: false

- GATT Server**

- Configurable: false

- Security Manager**

- Configurable: false

- System**

- Configurable: false

- **Bluetooth LE Controller (Link Layer)**

- Connection**

- Configurable: true

- Configs: sl_bluetooth_connection_config.h

- Legacy Advertising**

- Configurable: false

- Scanner for legacy advertisements**

- Configurable: false

- Platform

- **Board**

- Mainboard

- BRD4001A**

- Configurable: false

- Radio Board

BRD4180B

Configurable: false

- Bootloader

Bootloader Application Interface

Sources: btl_interface_ns.c, btl_interface_storage.c, btl_interface.c, btl_interface_nsc.c

Configurable: true

Configs: btl_interface_cfg_s2c4.h, btl_interface_cfg.h, btl_interface_cfg_s2c1.h,
btl_interface_cfg_s2c2.h, btl_interface_cfg_s2c3.h

- Device

- EFR32MG21

EFR32MG21A020F1024IM32

Sources: startup_efr32mg21.c, system_efr32mg21.c

Configurable: true

Configs: device.yaml

- Radio

RAIL Utility, PTI

Sources: sl_rail_util_pti.c

Configurable: true

Configs: sl_rail_util_pti_config.h

- Security

- Memory Protection Unit

Simple MPU

Sources: sl_mpu.c

Configurable: false

- Services

- Clock Manager

Clock Manager

Sources: sl_clock_manager_init.c

Configurable: true

Configs: sl_clock_manager_oscillator_config.h, sl_clock_manager_tree_config.h

- Device Initialization

Automatic Device Initialization

Configurable: false

- IO Stream

IO Stream: USART Core

Sources: sl_iostream_usart.c

Configurable: false

- Driver

IO Stream: USART

Configurable: true

Confgs: sl_iostream_usart_vcom_config.h

- System Setup

System Setup

Configurable: false

Pins:

Pin #	Pin Name	Function	Custom Pin Name	Software Component
17	PA00			
18	PA01			
19	PA02			
20	PA03	GPIO_SWV		SWO Debug
21	PA04	USART0_CTS		IO Stream: USART (vcom)
22	PA05	USART0_TX		IO Stream: USART (vcom)
23	PA06	USART0_RX		IO Stream: USART (vcom)
16	PB00			
15	PB01			
1	PC00			
2	PC01	USART0_RTS		IO Stream: USART (vcom)
3	PC02			
4	PC03			
5	PC04	PTI_DOUT		RAIL Utility, PTI
6	PC05	PTI_DFRAME		RAIL Utility, PTI
32	PD00			
31	PD01			
30	PD02			
29	PD03			
28	PD04	GPIO mode		Board Control

Peripherals:

Peripheral	Software Component	Custom Peripheral Name
ACMP0		
ACMP1		
CMU		
GPIO	SWO Debug	
I2C0		
I2C1		
IADC0		
LETIMER0		
LFXO		
MODEM		
PRS Channel ASYNCH0		

PRS Channel ASYNCH1		
PRS Channel ASYNCH10		
PRS Channel ASYNCH11		
PRS Channel ASYNCH2		
PRS Channel ASYNCH3		
PRS Channel ASYNCH4		
PRS Channel ASYNCH5		
PRS Channel ASYNCH6		
PRS Channel ASYNCH7		
PRS Channel ASYNCH8		
PRS Channel ASYNCH9		
PRS Channel SYNCH0		
PRS Channel SYNCH1		
PRS Channel SYNCH2		
PRS Channel SYNCH3		
PTI	RAIL Utility, PTI	
TIMER0		
TIMER1		
TIMER2		
TIMER3		
USART0	IO Stream: USART (vcom)	
USART1	IO Stream: USART (vcom)	
USART2	IO Stream: USART (vcom)	